

Problem

For the entire circuit in Fig. 11.70, calculate:

(a) the power factor

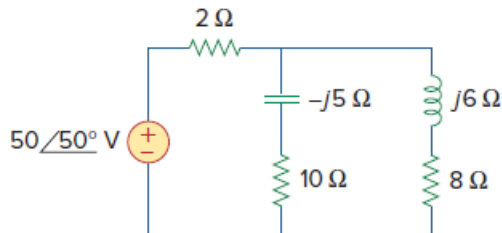
(b) the average power delivered by the source

(c) the reactive power

(d) the apparent power

(e) the complex power

Figure 11.70:



Step-by-step solution

Step 1 of 6

Refer to Figure 11.70 in the text book.

Calculate the value of the current I passing through the voltage source.

$$\begin{aligned}
 I &= \frac{50\angle 50^\circ}{2 + [(10 - j5) \parallel (8 + j6)]} \\
 &= \frac{50\angle 50^\circ}{2 + \left[\frac{(10 - j5)(8 + j6)}{(10 - j5) + (8 + j6)} \right]} \\
 &= \frac{50\angle 50^\circ}{2 + \left[\frac{110 + j20}{18 + j1} \right]} \\
 &= \frac{50\angle 50^\circ}{2 + 6.154 + j0.769} \\
 I &= \frac{50\angle 50^\circ}{8.154 + j0.769} \\
 &= \frac{50\angle 50^\circ}{8.19\angle 5.39^\circ} \\
 &= 6.105\angle 44.61^\circ \text{ A}
 \end{aligned}$$

Therefore, the value of the current I flowing through the voltage source is,

$$6.105\angle 44.61^\circ \text{ A}.$$

Step 2 of 6

(a)

Calculate the value of the power factor pf .

$$\text{pf} = \cos(\theta_v - \theta_i)$$

Substitute 50° for θ_v and 44.61° for θ_i .

$$\begin{aligned}\text{pf} &= \cos(50^\circ - 44.61^\circ) \\ &= \cos(5.39^\circ) \\ &= 0.9956\end{aligned}$$

Therefore, the value of the power factor pf is, 0.9956 .

Step 3 of 6

(b)

Calculate the value of the average power P .

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

Substitute 6.105 A for I_m , 50 V for V_m , 50° for θ_v and 44.61° for θ_i .

$$\begin{aligned}P &= \frac{1}{2} (50)(6.105) \cos(50^\circ - 44.61) \\ &= (152.625) \cos(5.39) \\ &= (152.625)(0.9956) \\ &= 151.95 \text{ W}\end{aligned}$$

Therefore, the value of the average power P is, 151.95 W .

Step 4 of 6

(c)

Calculate the value of the reactive power Q .

$$Q = \frac{1}{2} V_m I_m \sin(\theta_v - \theta_i)$$

Substitute 6.105 A for I_m , 50 V for V_m , 50° for θ_v and 44.61° for θ_i .

$$\begin{aligned}Q &= \frac{1}{2} (50)(6.105) \sin(50^\circ - 44.61) \\ &= (152.625) \sin(5.39) \\ &= (152.625)(0.0939) \\ &= 14.33 \text{ VAR}\end{aligned}$$

Therefore, the value of the reactive power Q is, 14.33 VAR .

Power factor is lagging since the value of the reactive power is positive.

Step 5 of 6

(d)

Calculate the value of the apparent power S .

$$S = \sqrt{P^2 + Q^2}$$

Substitute 14.33 VAR for Q and 151.95 W for P .

$$\begin{aligned} S &= \sqrt{(151.95)^2 + (14.33)^2} \\ &= \sqrt{23088.8 + 205.35} \\ &= \sqrt{23294.15} \\ &= 152.62 \text{ VA} \end{aligned}$$

Therefore, the value of the apparent power S is, 152.62 VA.

Step 6 of 6

(e)

Calculate the value complex power S .

$$S = \frac{1}{2} VI^*$$

Substitute $6.105 \angle 44.612^\circ$ A for I and $50 \angle 50^\circ$ V for V .

$$\begin{aligned} S &= \frac{1}{2} (50 \angle 50^\circ) (6.105 \angle 44.61^\circ)^* \\ &= (0.5) (50 \angle 50^\circ) (6.105 \angle -44.61^\circ) \\ &= 152.625 \angle 5.39^\circ \\ &= 151.95 + j14.34 \text{ VA} \end{aligned}$$

Therefore, the value of the complex power S is 151.95 + j14.34 VA.